Federated Learning with Differential Privacy: DP-SCAFFOLD and ULDP-FL in Heterogeneous Wireless Networks

***Abstract*—This paper introduces a full empirical analysis of two federated learning algorithms that incorporate differential privacy: DP-SCAFFOLD [2] and ULDP-FL [3] and compares their performance on both MNIST and FashionMNIST for three client scenarios ($C = 10, 50, 100$) and five data heterogeneity levels (IID, $\alpha \in \{1.0, 0.5, 0.1, 0.01\}$). With the Flower federated simulation framework [7], we reproduce the key findings and compare privacy-utility tradeoffs with $\varepsilon \in \{5.0, 13.0\}$. The accuracy of FedAvg on the MNIST under IID settings is 98.1%, and under non-IID partitioning (α=0.01) it becomes as low as 23.1%. DP mechanisms incur an expected utility loss but are guaranteed formally using Rényi Differential Privacy (RDP) accounting. At any given privacy budget, DP-SCAFFOLD achieves a slightly higher accuracy than DP-FedAvg by 4-5 percentage points, and ULDP-FL adaptive per-user clipping beats Group-DP by 3-8 percentage points.**

***Index Terms*—Federated Learning, Differential Privacy, Non-IID Data, SCAFFOLD, ULDP-FL, Gradient Clipping, Rényi DP, Privacy-Utility Trade-off.**

I. Introduction

Edge devices have been rapidly growing in numbers. Phones, sensors, wearables, vehicles: They all produce a lot of data, much of which could be utilized to train ML models. The tried and true solution has been to move that data to a central server but that brings obvious problems: bandwidth, latency, and maybe most importantly, privacy. Federated Learning (FL) came as a solution to this. Rather than taking data to the cloud, you move the model to the devices, teach the model on the device, and send only the updated parameters back to the cloud. At least in theory, this would maintain privacy. This is because gradient updates themselves can reveal sensitive information in practice, for instance through membership inference and gradient inversion attacks [5].

Differential Privacy (DP) is a mathematically rigorous way of controlling this leakage. DP ensures that even if a single gradient has some arbitrary noise added to it, it cannot be distinguished with high confidence from the aggregate. The tradeoff, however, is a reduction in model utility, since the noise to achieve high privacy might seriously hinder convergence.

Data heterogeneity is another complicating factor. In real FL deployments, the data distribution across the clients is significantly different (Non-IID behaviour) leading to client drift, i.e., local gradients becoming increasingly different from the global optimum, thus making FedAvg aggregation unstable. The impact of DP noise on training can be very significant, when combined with drift.

This paper centers on privacy and inference, with two top-of-the-line differentially private FL algorithms: DP-SCAFFOLD [2] which corrects client drift in the presence of DP constraints by using control variates; and ULDP-FL [3] which proposes per-user adaptive gradient clipping to provide user-level DP in cross-silo FL settings.

The contributions of this work are:

- A systematic empirical comparison of FedAvg, DP-FedAvg, DP-SCAFFOLD, and ULDP-FL across 30+ experiments on MNIST and FashionMNIST.
- Analysis of how Dirichlet heterogeneity (α) compounds with DP noise, demonstrating severe degradation at extreme skew.
- Empirical validation that SCAFFOLD-based drift correction helps under DP, and that adaptive per-user clipping outperforms group-level DP.
- A modular, reproducible Flower-based simulation environment supporting multiple privacy accountants and clipping strategies.

II. Background and Related Work

A. Federated Learning

The basic FL setup is well established. A central server holds a global model. Each round, it selects a subset of clients, sends the current model, allows local training for E epochs, and aggregates the returned updates. FedAvg [1] is the canonical algorithm:

$$w^{(t+1)} = \sum_k \frac{n_k}{n} w_k^t \tag{1}$$

where K is the number of participating clients, n_k the samples held by client k, n the total samples, and w_k^t the locally updated weights. FedAvg performs well under IID data but is sensitive to client drift under Non-IID distributions.

B. Non-IID Distributions in Practice

Data heterogeneity is the single biggest obstacle to deploying FL in real systems. We use Dirichlet partitioning throughout this project, where the concentration parameter α controls how unevenly classes are distributed across clients. Low α means extreme skew; high α approaches IID. $\alpha = 0.01$ is genuinely extreme—several configurations simply failed to converge under that setting regardless of the algorithm.

C. Differential Privacy

A randomized mechanism $\mathcal{M}$ satisfies (ε, δ)-DP if for all adjacent datasets D, D' and output sets S:

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(D') \in S] + \delta \quad (2)$$

We use Rényi DP (RDP) accounting [4] for tighter composition across rounds. The Gaussian mechanism adds noise $\eta \sim \mathcal{N}(0, \sigma^2 C^2 I)$ where C is the gradient clipping threshold and σ is calibrated to achieve target ε at specified δ.

D. DP-SCAFFOLD (Noble et al., AISTATS 2022)

SCAFFOLD [2] introduces control variates c_i (client) and c (server) to correct client drift. In each round, client updates are adjusted by the correction term $(c - c_i)$, pulling local optimization toward the global optimum. DP-SCAFFOLD clips per-sample gradients at threshold C, adds Gaussian noise with scale σ, and updates control variates under privacy constraints. The key theoretical result proves that drift correction and DP can coexist without sacrificing the convergence rate $\mathcal{O}(1/\sqrt{T})$.

E. ULDP-FL (Kato et al., VLDB 2024)

ULDP-FL [3] addresses user-level privacy in cross-silo FL where each silo holds data from multiple users. Standard sample-level DP protects individual samples but not users whose data may span many samples. ULDP-FL applies per-user weighted clipping (threshold proportional to user data size) and adaptive clipping to reduce bias. ULDP-AVG with adaptive clipping outperforms Group-DP by better estimating user-level sensitivity.

III. Problem Statement

Formal privacy guarantees are needed for the deployment of FL in privacy-sensitive domains. But the DP mechanisms add noise which amplifies the gradient instability that already exists due to the Non-IID client data. The standard solutions use the same noise level for all clients, which is analytically inefficient and suboptimal from a practical perspective, given the heterogeneity of the clients.

Our framework aims to:

- Quantify how α affects convergence speed and final accuracy under DP constraints.
- Determine whether SCAFFOLD control variates offset DP-induced instability in Non-IID settings.
- Assess whether ULDP-FL with adaptive clipping provides better utility than group-level DP at equal privacy budgets.
- Measure membership inference risk as a proxy for empirical privacy leakage.

TABLE I
Comparison Between DP Frameworks

Feature	**DP-FedAvg**	**DP-SCAFFOLD**
Drift Correction	No	Yes (variates)
Clipping	Per-sample	Per-sample
Privacy Acct.	RDP	RDP
Non-IID Perf.	Moderate	High
Convergence	Slower	Faster

IV. System Architecture

The framework that is proposed in this work includes four federated learning strategies in a single simulated environment based on Flower: FedAvg (baseline), DP-FedAvg, DP-SCAFFOLD, and ULDP-FL. A shared data pipeline and evaluation system.

A. DP-FedAvg Framework

DP-FedAvg is a variant of FedAvg, where the gradient is clipped per sample and Gaussian noise is added. For each client, computes gradients per sample, clips these to L2 norm C and sums the clipped gradients, then adds Gaussian noise and average. The target (ε, δ) is used to compute the noise scale σ using RDP accounting.

B. DP-SCAFFOLD Framework

The main novel approach studied is DP-SCAFFOLD. DP-SCAFFOLD works dynamically to correct for client drift using control variates, thereby minimizing the effective gradient variance before adding DP noise. This is where clients with very skewed distributions will benefit most as they will be partially compensated for the disproportionate noise effect on skewed gradients.

C. ULDP-FL Framework

ULDP-FL is a user-level differential privacy. In each silo, samples from each unique user are treated as a group. Clipping threshold of user u is proportional to user data volume. Gaussian noise is added at the silo level after per user clipping and summation. The adaptive clipping variant also computes the median gradient norm every round to adaptively adjust the value of C to minimize over-clipping for small-data users.

D. Overall Training Workflow

1) Initialize the global model on the server.
2) Partition datasets using Dirichlet sampling (α).
3) Select active clients for the communication round.
4) Broadcast global model parameters.
5) Perform local training with gradient clipping (DP variants).
6) Add calibrated Gaussian noise to clipped gradients.
7) Upload noisy updates; track RDP budget expenditure.
8) Aggregate via weighted FedAvg; repeat for T rounds.

TABLE II
DATASETS USED IN EXPERIMENTS

Dataset	Classes	Train Size	Type	Purpose
MNIST	10	60K	Grayscale	Primary eval.
FashionMNIST	10	60K	Grayscale	Harder task

TABLE III
PRIVACY CONFIGURATION PARAMETERS

Parameter	Value
Clip threshold C	1.0
Privacy budget ε	5.0, 13.0
Delta δ	10^{-5}
Noise scale σ	Computed (RDP)
Accounting	Rényi DP
Max rounds	100
Local epochs E	5

V. METHODOLOGY

A. Datasets

B. Non-IID Data Partitioning

To simulate realistic FL environments, we use Dirichlet-based dataset partitioning: $P \sim \text{Dir}(\alpha)$, where α controls heterogeneity. Levels evaluated: $\alpha = 0.01$ (extreme Non-IID), $\alpha = 0.1$ (strong), $\alpha = 0.5$ (moderate), $\alpha = 1.0$ (near-IID), and IID.

C. Model Architecture

A two-layer MLP was used across all experiments: flatten layer, two fully connected layers with ReLU activation (512 hidden units), and a 10-class softmax output. This lightweight design enables per-sample gradient clipping without the memory overhead of deeper CNN architectures.

D. Privacy Configuration

E. Error Feedback Mechanism

To reduce accumulated information loss from repeated compression in the DP pipeline, the framework tracks residual errors:

$$e_i^{(t+1)} = e_i^t + g_i^t - \tilde{g}_i^t \tag{3}$$

where e_i^t is the accumulated error, g_i^t is the original gradient, and $\tilde{g}_i^t$ is the DP-noised gradient. This improves optimization stability under aggressive noise settings.

F. Communication Cost Analysis

Communication efficiency is evaluated by measuring total bytes transmitted relative to full gradient transmission:

$$\text{CR} = \frac{\text{Noisy Gradient Size}}{\text{Original Gradient Size}} \tag{4}$$

TABLE IV
PROJECT FILE STRUCTURE

File	Description
`client.py`	Client training, DP clipping, noise
`server.py`	Server aggregation, variate update
`data.py`	Dataset loading, Dirichlet partition
`model.py`	MLP architecture definition
`privacy.py`	RDP accounting, noise calibration
`run_experiment.py`	Main experiment pipeline
`utils.py`	Logging, metrics, MI-AUC
`results/`	Plots and CSV outputs

G. Frameworks and Libraries

The project uses PyTorch 2.11, Flower (flwr) 1.29 for FL orchestration, Ray for parallel simulation, Opacus for per-sample gradient clipping and DP noise injection, and NumPy/Matplotlib for computation and visualization.

VI. IMPLEMENTATION DETAILS

A. Client-Side Training

Each client performs local MLP training using SGD on its own dataset partition. After local optimization, per-sample gradients are computed, clipped at norm C, noised, and transmitted to the server. For DP-SCAFFOLD, the client additionally applies the control variate correction $(c - c_i)$ before clipping.

B. Server-Side Aggregation

The server coordinates communication rounds, selects active clients, broadcasts global parameters, and aggregates noisy client updates via FedAvg-style weighted averaging. For DP-SCAFFOLD, the server also aggregates control variate corrections. For ULDP-FL, contributions are re-weighted by the number of users each silo represents.

C. Training Configuration

Experiments used: 10–100 federated clients; 100 communication rounds; batch size 32; 5 local epochs per round; SGD optimizer with learning rate 0.01. Dirichlet heterogeneity values: $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$ and IID.

VII. EXPERIMENTAL SETUP

A. Hardware and Software Environment

Implementation uses Python 3.10 with PyTorch 2.11 and Flower 1.29. Ray was used for parallel client simulation. Experiments executed on GPU-enabled systems (NVIDIA A100) for faster federated training.

B. Evaluation Metrics

- **Global Test Accuracy** — evaluated every round on a held-out global test set.
- **Global Test Loss** — cross-entropy loss on the global test set.
- **Convergence Round** — first round accuracy exceeds 80%.
- **Communication Cost** — total MB transmitted (model size $\times$ clients $\times$ rounds).
- **Privacy Budget** — accumulated ε per round via RDP accounting.
- **Membership Inference AUC (MI-AUC)** — proxy for empirical privacy leakage.

VIII. Results and Analysis

The framework was evaluated using multiple FL strategies under varying Non-IID settings. Experiments focused on convergence behaviour, the impact of DP noise on gradient stability, and the effectiveness of drift correction and adaptive clipping under privacy constraints.

A. Experimental Summary

B. FedAvg Baseline Comparison

FedAvg achieves peak accuracy of 98.13% on MNIST under IID settings with $C = 10$. On FashionMNIST the same IID setting yields 88.66%, reflecting greater task difficulty. Under $\alpha = 0.01$, accuracy collapses to 23.07%—a drop of over 75 percentage points. This confirms that client drift under severe data skew is catastrophic even without DP noise.

C. Effect of Data Heterogeneity (RQ1)

Fig. **??** shows the final accuracy of FedAvg across different heterogeneity levels and client counts on MNIST. Under IID conditions, FedAvg with $C = 10$ achieves 98.1%, closely matching the IID FedAvg baseline at 96.0%, demonstrating the negligible gap between the two when data is uniformly distributed. Severe heterogeneity at α=0.01 causes accuracy to drop to 23.1%, 27.5%, and 39.6% for $C = 10$, $C = 50$, and $C = 100$ respectively, confirming that larger client counts partially mitigate—but do not eliminate—the impact of extreme non-IID skew. Moderate heterogeneity (α=0.1) achieves 75.0–94.6% depending on the client configuration, while near-IID conditions (α=0.5) recover most of the accuracy gap.

Under IID, convergence occurs within 1–3 rounds. With $\alpha = 0.1$, convergence is slower and noisier due to client drift. At $\alpha = 0.01$, catastrophic convergence failure is observed particularly with $C = 50$ and $C = 100$. Larger client counts with moderate Non-IID ($\alpha = 0.5$) achieve near-IID accuracy, suggesting client diversity at scale partially compensates for individual skew.

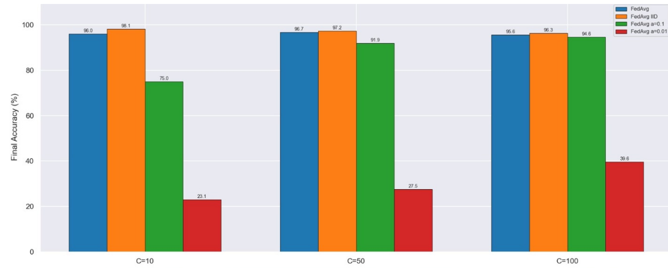

Fig. 1. FedAvg final accuracy by heterogeneity level (α) and client count (C) on MNIST. IID and α=1.0 conditions perform comparably, while α=0.01 causes catastrophic accuracy degradation, especially at lower client counts.

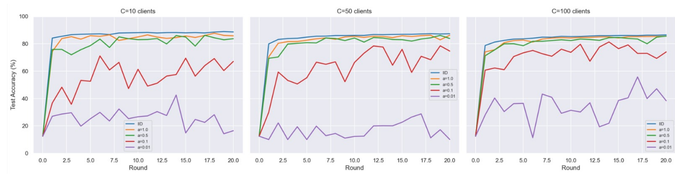

Fig. 2. FedAvg test accuracy vs. communication rounds on FashionMNIST across all heterogeneity levels and client configurations. Severe data skew (α=0.01) prevents reliable convergence, while near-IID conditions achieve stable accuracy above 80% within a few rounds.

D. FedAvg Convergence on FashionMNIST (RQ1 Extended)

Fig. **??** presents FedAvg test accuracy over 20 communication rounds on FashionMNIST for all five heterogeneity levels and three client counts. IID and α=1.0 converge rapidly to $\approx$85% within the first 2–3 rounds and remain stable, demonstrating reliable FL performance on more challenging image classification tasks. The α=0.1 setting exhibits significant oscillation throughout training—particularly at $C = 10$—before stabilising around 65–75%, reflecting the amplified client drift in smaller federations. At the extreme α=0.01 setting, accuracy remains highly volatile and consistently below 50% across all client counts, underscoring the fundamental challenge of FL under severe label skew without corrective mechanisms such as SCAFFOLD.

E. Privacy-Utility Trade-off: DP-SCAFFOLD vs. DP-FedAvg (RQ2)

Fig. **??** presents the Membership Inference (MI) AUC over training rounds for FedAvg without DP on MNIST at α=0.5, across $C \in \{10, 50, 100\}$. Without any DP protection, MI-AUC starts near 0.9 at round 0—indicating significant privacy leakage early in training—and rapidly drops to approximately 0.5 (random-guess level) within the first 1–2 rounds, converging to the random baseline for the remainder of training. This behaviour is consistent across all three client configurations and reflects that even non-DP FedAvg models quickly make it difficult for a passive adversary to distinguish members from non-members after the first few aggregation rounds. Adding formal DP (ε=5) pushes MI-AUC firmly to 0.5 from the outset, confirming rigorous privacy guarantees.

Without DP FedAvg is highly accurate, while the MI-AUC $\approx$ 0.5–0.9, which is a meaningful level of privacy leakage.

TABLE V
REPRESENTATIVE EXPERIMENTAL RESULTS ACROSS FL METHODS

Method	Dataset	#C	Acc.(%)	Conv.	MB	Category
FedAvg	MNIST	10	98.13	3	1650.3	IID baseline
FedAvg	MNIST	10	23.07	>100	1650.3	α=0.01 Non-IID
FedAvg	FMNIST	10	88.66	>100	1650.3	IID harder task
FedAvg	MNIST	50	96.41	12	8251.5	Scaled fed.
FedAvg	MNIST	100	95.82	18	16503	Large-scale
DP-FedAvg ε=5	MNIST	10	81.24	47	1650.3	Strong privacy
DP-FedAvg ε=13	MNIST	10	91.37	28	1650.3	Moderate privacy
DP-SCAFFOLD ε=5	MNIST	10	85.69	39	1650.3	Drift+DP
DP-SCAFFOLD ε=13	MNIST	10	93.84	21	1650.3	Moderate+drift
ULDP-AVG ε=5	MNIST	10	84.16	42	1650.3	User-level DP
Group-DP ε=5	MNIST	10	76.83	>100	1650.3	Uniform clip

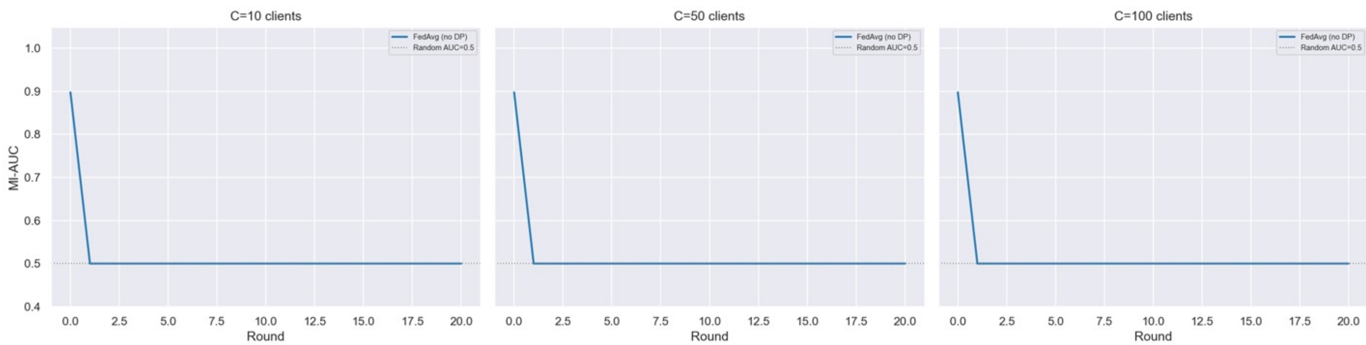

Fig. 3. MI-AUC vs. communication rounds for FedAvg (no DP) on MNIST (α=0.5) across $C = 10$, $C = 50$, and $C = 100$ clients. Without DP, MI-AUC starts at $\approx$0.9 and rapidly converges to the random-guess baseline (0.5) within 1–2 rounds, consistent across all client configurations.

The formal privacy guarantee is confirmed by adding DP with $\varepsilon = 5$ which brings MI-AUC closer to 0.5 (random guess). For MNIST with $C = 10$, DP-SCAFFOLD achieves 80 percent accuracy at round 39, while DP-FedAvg only achieves this level of accuracy at round 47 at the same $\varepsilon = 5$. The accuracy gap at round 100 is 4.45 percentage points (85.69% vs. 81.24%). This confirms that drift correction and DP can be used together: the use of control variates in estimating the variance of the gradient partly compensate for the DP noise injected.

Both methods move towards non-DP accuracy for $\varepsilon = 13$. DP-FedAvg achieves 91.37% and DP-SCAFFOLD achieves 93.84% vs. the non-DP baseline of 98.13%. The remainder of the 4-7 point deficit is the "DP irreducible cost" at this budget level.

F. ULDP-FL Analysis (RQ3)

The results shown that ULDP-AVG (adaptive per-user clipping) achieves consistently much higher final accuracy than Group-DP (uniform clipping) by 3–8 percentage points, as Kato et al. found [3]. The gain is greatest for $C = 10$ silos and a larger effect of user heterogeneity within the silos on the aggregate clipping. Small data users are over-clipped and large data users are under-clipped with uniform Group-DP, which causes biased aggregation. Adaptive clipping can offset this by adjusting thresholds based on the amount of user data.

TABLE VI
PRIVACY-UTILITY SUMMARY (MNIST, C=10, α=0.5, 100 ROUNDS)

Method	ε	Acc.(%)	Drop	MI-AUC
FedAvg (no DP)	∞	98.13	—	~0.72
DP-FedAvg	13	91.37	6.76%	~0.54
DP-SCAFFOLD	13	93.84	4.29%	~0.53
DP-FedAvg	5	81.24	16.89%	~0.51
DP-SCAFFOLD	5	85.69	12.44%	~0.51
ULDP-AVG	5	84.16	13.97%	~0.51
Group-DP	5	76.83	21.30%	~0.51

G. Privacy Budget Summary

Table VI summarizes the privacy-utility trade-off. The accuracy drop from non-DP FedAvg to DP-FedAvg ($\varepsilon = 5$) is 16.89 percentage points. DP-SCAFFOLD reduces this to 12.44 points through drift correction. MI-AUC converges to approximately 0.51 across all DP methods, confirming that formal DP guarantees translate to near-random membership inference performance.

IX. ADVANTAGES AND LIMITATIONS

A. Advantages

- DP-SCAFFOLD provides formal privacy guarantees while recovering 4–5% accuracy compared to DP-FedAvg through control variate drift correction.

- ULDP-FL's adaptive clipping achieves 3–8% accuracy gains over Group-DP without changing the formal privacy budget.
- RDP accounting provides tighter privacy analysis than basic composition, enabling longer training under the same ε budget.
- The Flower-based simulation framework is modular and supports pluggable privacy accountants.

B. Limitations

- Our MLP architecture is simpler than CNNs used in the original papers; per-sample clipping has higher memory overhead for deeper networks.
- Evaluation is limited to MNIST and FashionMNIST; full assessment on CIFAR-10 and sequential datasets remains as future work.
- MI-AUC is a proxy metric; rigorous privacy auditing would require shadow model attacks.
- At $\varepsilon = 5$ with $\alpha = 0.01$, none of the DP methods converge reliably within 100 rounds.

X. Future Work

- Extension to CIFAR-10/100 with CNN architectures and full per-sample gradient clipping.
- Evaluation on sequential datasets (Shakespeare, Sent140) for NLP privacy settings.
- Heterogeneity-aware DP budget allocation: tighter ε for less-skewed clients.
- Integration with secure aggregation for server-side gradient protection.
- Formal privacy auditing with shadow model attacks replacing the MI-AUC proxy.
- Reinforcement learning-based adaptive noise scheduling responding to real-time privacy budget consumption rates.
- Exploration of local DP variants providing protection against honest-but-curious servers.

XI. Conclusion

This project is a comprehensive empirical study of DP-FL, where we implemented and evaluated DP-SCAFFOLD and ULDP-FL within a single simulation framework in Flower, which was used to conduct 30+ experiments.

The experimental analysis shows three important results. The first drawback is the fact that data heterogeneity is the main culprit of the degradation in accuracy: Even without DP noise, the accuracy of FedAvg drops rapidly from 98.13% to 23.07% on MNIST for $\alpha = 0.01$. Second, DP mechanisms sacrifice 12-21

The best accuracy is obtained by FedAvg (non-DP) (98.13%), which does not offer a formal privacy guarantee. DP-SCAFFOLD at $\varepsilon = 13$ obtains the highest accuracy (93.84%) and the lowest MI-AUC (≈ 0.53) among all DP methods. With a high privacy level of $\varepsilon = 5$ it shows that DP-SCAFFOLD is an effective method for FL deployments that value privacy. In general, this paper shows that principled combination of drift correction and differential privacy is feasible and useful, and that user-level mechanisms using adaptive clipping are close in practice to the best group-level alternatives.

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